Customer Shopping Behavior Analysis

A comprehensive exploration of transactional data from 3,900 purchases to uncover strategic insights into customer segments, product preferences, and revenue opportunities.

Project Overview

Objective

Analyze transactional data to uncover spending patterns, customer segments, and product preferences that guide strategic business decisions.

Scope

3,900 purchases across product categories with 18 data points including demographics, purchase details, and shopping behavior.

Approach

Python for data preparation and cleaning, PostgreSQL for structured analysis, and Power BI for interactive visualization.

Dataset Summary

3,900

Purchase Records

Complete transaction history analyzed

18

Data Columns

Rich feature set captured

37

Missing Values

Addressed in review ratings

Key Feature Categories

Customer Demographics

Age, Gender, Location, and Subscription Status

Purchase Details

Item Purchased, Category, Purchase Amount, Season, Size, Color

Shopping Behavior

Discount Applied, Promo Code Used, Previous Purchases, Frequency, Review Rating, Shipping Type

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Data Preparation & Cleaning

Python was used to load, explore, and clean the dataset before PostgreSQL analysis.

Data Loading
Imported dataset using pandas with initial structure check via df.info()

Exploration
Generated summary statistics using .describe() for numerical features

Redundancy Check
Verified discount_applied and promo_code_used were redundant; dropped promo_code_used

Missing Data
Imputed 37 missing review ratings using median per product category

Standardization
Rename columns to snake_case for better readability

Feature Engineering
Created age_group bins and purchase_frequency_days from raw data

Database Load
Connected to PostgreSQL and loaded cleaned DataFrame

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Key SQL Analysis Results

Revenue by Gender

Male customers generated $157,890 versus Female customers at $75,191

Shipping Comparison

Express shipping average: $60.48 vs. Standard: $58.46

Top Products

Top 3 per category identified—Blouse and Jewelry tied at 171 orders

839

High-Spending Discount Users

Customers using discounts but spending above average purchase amount of $59.76

50%

Highest Discount Rate

Hats received discounts in 50% of purchases—highest among all products

3.86

Top Product Rating

Gloves achieved highest average rating at 3.86 out of 5 stars

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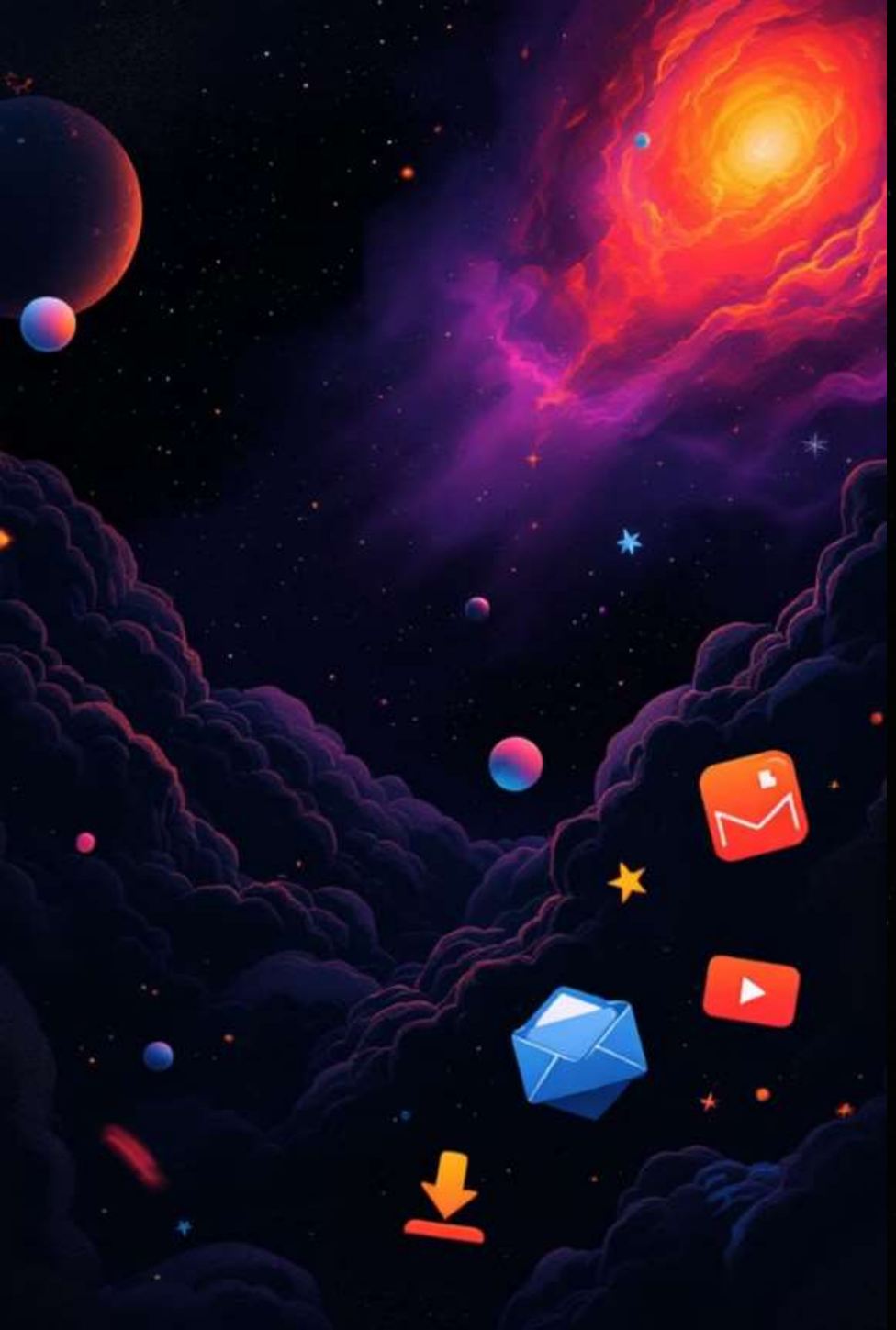

Subscription & Customer Segmentation

Subscription Performance

Non-subscribers generated higher total revenue ($170,436) with 2,847 customers, while subscribers averaged slightly lower per-customer spend ($59.49 vs. $59.87) with 1,053 customers.

Customer Segments

Segmented by purchase history: Loyal (3,116 customers), Returning (701), and New (83). Loyal customers represent 79.9% of the customer base.

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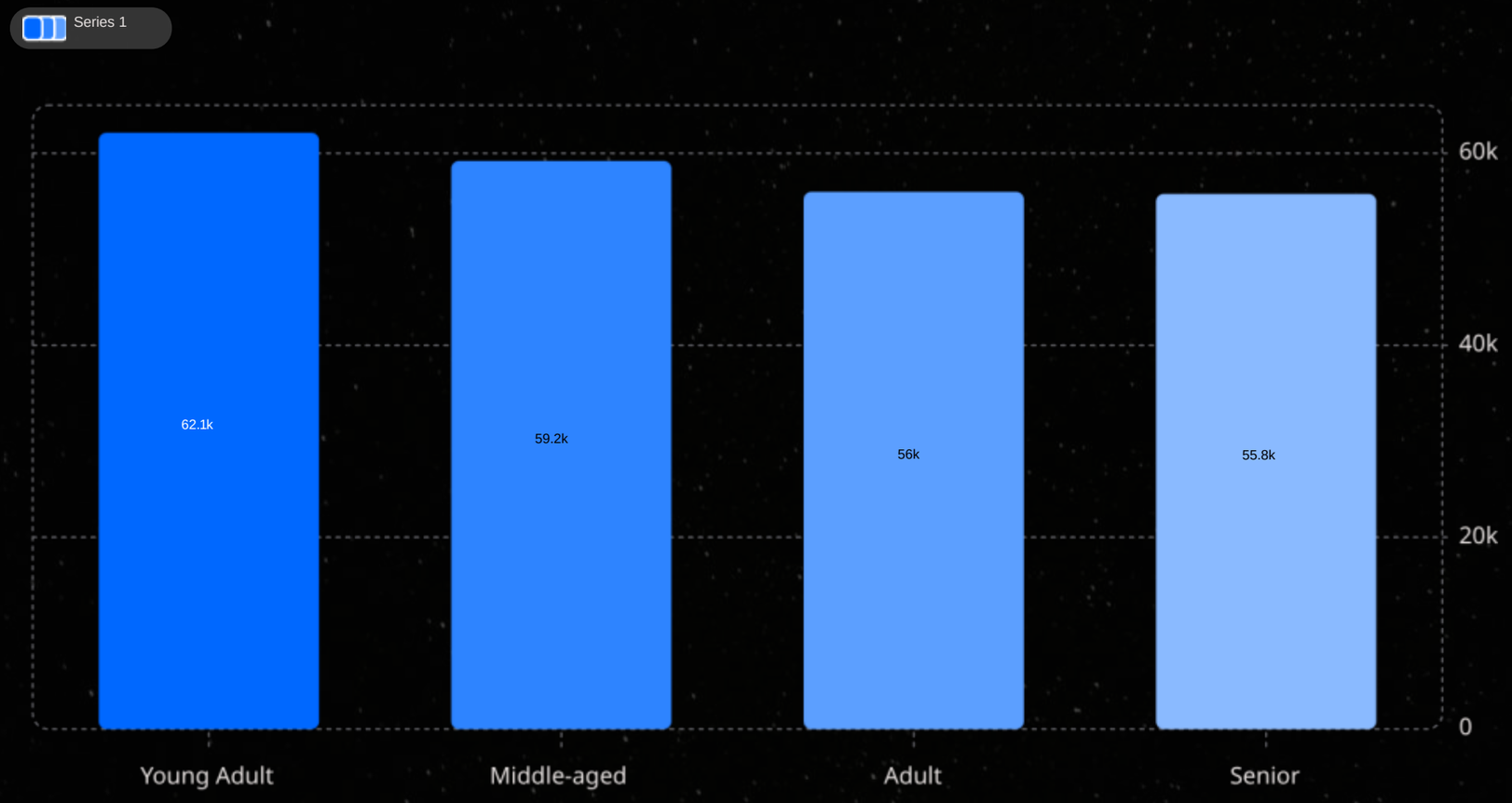
Age Group Revenue Analysis

Total revenue contribution across four age segments analyzed for targeted marketing opportunities.



Young Adults generated the highest revenue at $62,143, followed by Middle-aged customers at $59,197. Adult and Senior segments showed similar performance with $55,978 and $55,763 respectively.

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Power BI Dashboard

KPIs

- 3,900 customers
- $59.76 average purchase
- 3.75 average rating

Filters

- Subscription status
- Gender
- Category
- Shipping type

Visualizations

- Subscription status donut
- Revenue by category
- Sales by age group

Business Recommendations

1

Boost Subscriptions

Promote exclusive benefits and incentives to convert non-subscribers. Current base at 27% offers significant growth opportunity.

2

Enhance Loyalty Programs

Reward returning buyers to accelerate movement into loyal segment where 79.9% of customers already reside.

3

Optimize Discount Strategy

Balance sales volume with margin control. High discount rates (50% on hats) may erode profitability.

4

Highlight Top Products

Feature best-selling items (Blouse, Pants at 171 orders) and top-rated products (Gloves at 3.86 stars) in campaigns.

5

Target High-Value Segments

Focus marketing on Young Adult and Middle-aged demographics generating highest revenue. Prioritize express-shipping customers.

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Next Steps

Immediate Actions

- Launch subscription promotion campaign within 30 days
- Develop loyalty tier structure with exclusive benefits
- Create targeted email sequences for age segments
- Review discount thresholds with finance team

Ongoing Analysis

- Monitor subscription conversion rates monthly
- Track repeat purchase frequency trends
- Analyze seasonal purchase patterns
- Assess product rating correlations with revenue

> "Data-driven insights transform customer understanding into competitive advantage."

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